🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.DS.5

Display Data: Box Plots

Lesson 8-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can make and read a box plot to summarize a data set.

Language Objective: I can explain my box plot using the words box plot, median, quartile, and interquartile range.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Box Plot | **Median** | **Quartile** | **Interquartile Range** | **Data distribution** | **Variability**

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A graph that shows data with a box and whiskers based on the five-number summary is a ____________ .

2. The middle value of the data, shown as a line inside the box, is the ____________ .

3. A value that divides the data into four equal parts is a ____________ .

4. The distance between the first and third quartiles, showing the middle 50% of data, is the ____________ .

5. The way data values are spread out is the ____________ .

6 How spread out the data values are is the ________________.

Reflect — Exit Ticket

A box plot has Q1 = 20 and Q3 = 36. What is the IQR and what does it represent?

A. IQR = 16; it is the spread of the middle 50% of the data

B. IQR = 56; it is the total of Q1 and Q3

C. IQR = 28; it is the median between Q1 and Q3

D. IQR = 36; it is the value of Q3

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Box Plot
2. **Notes 2:** Median
3. **Notes 3:** Quartile
4. **Notes 4:** Interquartile Range
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Display Data: Box Plots is skipping the key idea: "The box always holds the middle 50% of the data, and the line inside the box is the median." — always check your work against this rule before you submit.*
8. **Try It 1:** A. Q1 = 12, Q3 = 28 — *With 8 values the median splits the data into a lower half (6, 10, 14, 18) and an upper half (22, 26, 30, 34). Q1 is the median of the lower half: (10 + 14) ÷ 2 = 12. Q3 is the median of the upper half: (26 + 30) ÷ 2 = 28.*
9. **Try It 2:** A. Between 15 and 30 — *The box of a box plot stretches from Q1 to Q3, and that box always holds the middle 50% of the data. Here the box goes from Q1 = 15 to Q3 = 30.*
10. **Exit Ticket:** A. IQR = 16; it is the spread of the middle 50% of the data — *IQR = Q3 − Q1 = 36 − 20 = 16. The IQR tells you the range of the middle 50% of the data.*